

6 Testing and Performance Analysis

6.1 Testbed Validation

Successful deployment and operation of the O-RAN compliant 5G Standalone testbed were verified through the execution of the 5G Core Network, FlexRIC Near-RT RIC, OAI gNB, and OAI NR UE. The successful initialization of all core network functions and RIC services is shown in the deployment screenshots.

The gNB and UE logs confirmed successful:

- RRC connection establishment
- PDU session setup
- UE synchronization
- MAC layer scheduling
- GTP tunnel establishment

The corresponding screenshots of the gNB and UE operational states are shown in the implementation outputs.

6.2 UDP Throughput Analysis

UDP throughput testing was performed using `iperf3` across both uplink (UE → Core) and downlink (Core → UE) directions for multiple target bitrates ranging from 20 Mbps to 50 Mbps. The test duration for each experiment was set to 10 seconds.

6.2.1 UDP Uplink Performance

The UDP uplink test results demonstrated stable operation up to a target bitrate of 44 Mbps. Uplink throughput test data for 20 Mbps, 30 Mbps, 40 Mbps, and 44 Mbps are shown in the table.

Screenshots of uplink throughput test of 44 Mbps is shown.

```
etccomm@etccomm-Veriton-Series:~/O-RAN-Testbed-Automation/OpenAirInterface_Testbed$ sudo ip netns exec ue1 iperf3 -c 10.45.0.1 -B 10.45.0.2 -u -b 44M
Connecting to host 10.45.0.1, port 5201
[ 5] local 10.45.0.2 port 52479 connected to 10.45.0.1 port 5201
[ ID] Interval      Transfer    Bitrate    Total Datagrams
[ 5] 0.00-1.00 sec  5.24 MBytes  44.0 Mbits/sec  3797
[ 5] 1.00-2.00 sec  5.23 MBytes  43.8 Mbits/sec  3790
[ 5] 2.00-3.00 sec  5.25 MBytes  44.1 Mbits/sec  3805
[ 5] 3.00-4.00 sec  5.24 MBytes  43.8 Mbits/sec  3792
[ 5] 4.00-5.00 sec  5.25 MBytes  44.2 Mbits/sec  3805
[ 5] 5.00-6.00 sec  5.24 MBytes  44.0 Mbits/sec  3798
[ 5] 6.00-7.00 sec  5.25 MBytes  44.0 Mbits/sec  3799
[ 5] 7.00-8.00 sec  5.24 MBytes  44.0 Mbits/sec  3795
[ 5] 8.00-9.00 sec  5.25 MBytes  44.1 Mbits/sec  3802
[ 5] 9.00-10.00 sec 5.23 MBytes  43.9 Mbits/sec  3790
- - - - -
[ ID] Interval      Transfer    Bitrate    Jitter    Lost/Total Datagrams
[ 5] 0.00-10.00 sec  52.4 MBytes  44.0 Mbits/sec  0.000 ms  0/37973 (0%) sender
[ 5] 0.00-48.77 sec  52.4 MBytes  9.02 Mbits/sec  1.041 ms  0/37973 (0%) receiver
iperf Done.
etccomm@etccomm-Veriton-Series:~/O-RAN-Testbed-Automation/OpenAirInterface_Testbed$
```

```

Server listening on 5201
-----
Accepted connection from 10.45.0.2, port 60221
[ S] local 10.45.0.1 port 5201 connected to 10.45.0.2 port 52479
[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Total Datagrams
[ S] 0.00-1.00 sec    178 KBytes    1.45 Mbits/sec  5.388 ms    0/126 (0%)
[ S] 1.00-2.00 sec    331 KBytes    2.72 Mbits/sec  4.145 ms    0/234 (0%)
[ S] 2.00-3.00 sec    465 KBytes    3.82 Mbits/sec  2.238 ms    0/329 (0%)
[ S] 3.00-4.00 sec    591 KBytes    4.85 Mbits/sec  2.165 ms    0/418 (0%)
[ S] 4.00-5.01 sec    684 KBytes    5.57 Mbits/sec  3.095 ms    0/484 (0%)
[ S] 5.01-6.00 sec    936 KBytes    7.71 Mbits/sec  1.822 ms    0/662 (0%)
[ S] 6.00-7.00 sec    1.10 MBytes   9.26 Mbits/sec  1.509 ms    0/799 (0%)
[ S] 7.00-8.00 sec    1.28 MBytes   10.7 Mbits/sec  0.776 ms    0/927 (0%)
[ S] 8.00-9.00 sec    1.28 MBytes   10.8 Mbits/sec  0.579 ms    0/929 (0%)
[ S] 9.00-10.00 sec   1.26 MBytes   10.6 Mbits/sec  0.798 ms    0/913 (0%)
[ S] 10.00-11.00 sec  1.16 MBytes   9.72 Mbits/sec  0.591 ms    0/839 (0%)
[ S] 11.00-12.01 sec  1.13 MBytes   9.40 Mbits/sec  3.280 ms    0/818 (0%)
[ S] 12.01-13.00 sec  1.19 MBytes   10.0 Mbits/sec  0.906 ms    0/859 (0%)
[ S] 13.00-14.00 sec  1.15 MBytes   9.65 Mbits/sec  1.212 ms    0/833 (0%)
[ S] 14.00-15.00 sec  1.15 MBytes   9.61 Mbits/sec  0.988 ms    0/830 (0%)
[ S] 15.00-16.00 sec  1.14 MBytes   9.55 Mbits/sec  1.040 ms    0/824 (0%)
[ S] 16.00-17.00 sec  1.14 MBytes   9.53 Mbits/sec  1.440 ms    0/824 (0%)
[ S] 17.00-18.00 sec  1.14 MBytes   9.55 Mbits/sec  1.000 ms    0/823 (0%)
[ S] 18.00-19.00 sec  1.16 MBytes   9.72 Mbits/sec  1.645 ms    0/840 (0%)
[ S] 19.00-20.00 sec  1.14 MBytes   9.55 Mbits/sec  1.655 ms    0/824 (0%)
[ S] 20.00-21.00 sec  1.11 MBytes   9.35 Mbits/sec  1.017 ms    0/807 (0%)
[ S] 21.00-22.00 sec  1.16 MBytes   9.70 Mbits/sec  1.605 ms    0/840 (0%)
[ S] 22.00-23.00 sec  1.16 MBytes   9.73 Mbits/sec  1.025 ms    0/837 (0%)
[ S] 23.00-24.00 sec  1.14 MBytes   9.53 Mbits/sec  1.446 ms    0/823 (0%)
[ S] 24.00-25.00 sec  1.13 MBytes   9.49 Mbits/sec  1.395 ms    0/819 (0%)
[ S] 25.00-26.06 sec  1.14 MBytes   9.03 Mbits/sec  0.902 ms    0/829 (0%)
[ S] 26.06-27.00 sec  1.17 MBytes   10.4 Mbits/sec  1.506 ms    0/845 (0%)
[ S] 27.00-28.00 sec  1.11 MBytes   9.31 Mbits/sec  1.119 ms    0/804 (0%)
[ S] 28.00-29.00 sec  1.14 MBytes   9.53 Mbits/sec  0.907 ms    0/823 (0%)
[ S] 29.00-30.00 sec  1.16 MBytes   9.72 Mbits/sec  0.867 ms    0/839 (0%)
[ S] 30.00-31.00 sec  1.14 MBytes   9.53 Mbits/sec  1.142 ms    0/823 (0%)
[ S] 31.00-32.00 sec  1.12 MBytes   9.36 Mbits/sec  0.929 ms    0/808 (0%)
[ S] 32.00-33.00 sec  1.12 MBytes   9.40 Mbits/sec  1.859 ms    0/812 (0%)
[ S] 33.00-34.00 sec  1.18 MBytes   9.86 Mbits/sec  0.760 ms    0/851 (0%)
[ S] 34.00-35.00 sec  1.14 MBytes   9.55 Mbits/sec  1.688 ms    0/825 (0%)
[ S] 35.00-36.00 sec  1.18 MBytes   9.89 Mbits/sec  0.922 ms    0/853 (0%)
[ S] 36.00-37.00 sec  1.14 MBytes   9.53 Mbits/sec  1.662 ms    0/824 (0%)
[ S] 37.00-38.03 sec  1.14 MBytes   9.27 Mbits/sec  3.248 ms    0/824 (0%)
[ S] 38.03-39.00 sec  1.16 MBytes   10.0 Mbits/sec  1.047 ms    0/837 (0%)
[ S] 39.00-40.00 sec  1.16 MBytes   9.72 Mbits/sec  0.854 ms    0/839 (0%)
[ S] 40.00-41.00 sec  1.12 MBytes   9.38 Mbits/sec  2.098 ms    0/811 (0%)
[ S] 41.00-42.00 sec  1.16 MBytes   9.71 Mbits/sec  0.933 ms    0/837 (0%)
[ S] 42.00-43.00 sec  1.14 MBytes   9.55 Mbits/sec  0.847 ms    0/824 (0%)
[ S] 43.00-44.00 sec  1.14 MBytes   9.53 Mbits/sec  1.010 ms    0/823 (0%)
[ S] 44.00-45.00 sec  1.10 MBytes   9.21 Mbits/sec  2.850 ms    0/795 (0%)
[ S] 45.00-46.00 sec  1.18 MBytes   9.88 Mbits/sec  1.557 ms    0/854 (0%)
[ S] 46.00-47.00 sec  1.18 MBytes   9.80 Mbits/sec  1.280 ms    0/852 (0%)
[ S] 47.00-48.00 sec  1.14 MBytes   9.55 Mbits/sec  2.019 ms    0/825 (0%)
[ S] 48.00-48.77 sec  926 KBytes    9.86 Mbits/sec  1.041 ms    0/655 (0%)
-----
[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Total Datagrams
[ S] 0.00-48.77 sec  52.4 MBytes   9.02 Mbits/sec  1.041 ms    0/37973 (0%) receiver

```

Figure 8: 44Mbps UDP uplink Iperf3 test

Test Duration set at 10 secs

Table 4: UDP Uplink Performance (UE → Core)

Target Rate (Mbps)	Sending Bit Rate (Mbps)	Data Transferred (MB)	Receiving Bit Rate (Mbps)	Data Received (MB)	Arrival Time (s)	Loss (%)	Status
20	20	23.8	8.34	23.8	23.96	0	Did not crash
30	30	35.8	8.65	35.8	34.66	0	Did not crash
40	40	47.7	8.90	47.7	44.92	0	Did not crash
44	44	52.4	9.02	52.4	48.77	0	Did not crash
45	45	Crashed at 9 seconds					
50	50	Crashed at 9 seconds					

At 45 Mbps and above, the iperf3 control socket closed unexpectedly, terminating the test prematurely. The screenshots corresponding to the 45 Mbps and 50 Mbps failures clearly show abrupt socket closure and bad file descriptor errors.

```

etccomm@etccomm-Veriton-Series:~/0-RAN-Testbed-Automation/OpenAirInterface_Testbed$ sudo ip netns exec ue1 iperf3 -c 10.45.0.1 -B 10.45.0.2 -u -b 45M
Connecting to host 10.45.0.1, port 5201
[ 5] local 10.45.0.2 port 35613 connected to 10.45.0.1 port 5201
[ ID] Interval      Transfer      Bitrate      Total Datagrams
[ 5] 0.00-1.00 sec  5.36 MBytes  45.0 Mbits/sec  3885
[ 5] 1.00-2.00 sec  5.36 MBytes  44.9 Mbits/sec  3881
[ 5] 2.00-3.00 sec  5.34 MBytes  44.8 Mbits/sec  3865
[ 5] 3.00-4.00 sec  5.40 MBytes  45.3 Mbits/sec  3907
[ 5] 4.00-5.01 sec  5.34 MBytes  44.4 Mbits/sec  3869
[ 5] 5.01-6.00 sec  5.37 MBytes  45.5 Mbits/sec  3891
[ 5] 6.00-7.00 sec  5.37 MBytes  45.1 Mbits/sec  3892
[ 5] 7.00-8.00 sec  5.36 MBytes  45.0 Mbits/sec  3884
[ 5] 8.00-9.00 sec  5.36 MBytes  45.0 Mbits/sec  3885
iperf3: error - control socket has closed unexpectedly

```

Figure 9: 45Mbps UDP Iperf3 test failure

6.2.2 UDP Downlink Performance

UDP downlink testing exhibited better throughput performance compared to uplink due to the asymmetric TDD slot allocation favoring downlink transmission.

```

etccomm@etccomm-Veriton-Series:~/0-RAN-Testbed-Automation/OpenAirInterface_Testbed$ sudo ip netns exec ue1 iperf3 -c 10.45.0.1 -B 10.45.0.2 -u -b 44M
-R
Connecting to host 10.45.0.1, port 5201
Reverse mode, remote host 10.45.0.1 is sending
[ 5] local 10.45.0.2 port 44680 connected to 10.45.0.1 port 5201
[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Total Datagrams
[ 5] 0.00-1.00 sec  1.31 MBytes  11.0 Mbits/sec  0.943 ms  0/946 (0%)
[ 5] 1.00-2.02 sec  1.55 MBytes  12.7 Mbits/sec  1.584 ms  0/1126 (0%)
[ 5] 2.02-3.00 sec  1.91 MBytes  16.5 Mbits/sec  0.517 ms  0/1386 (0%)
[ 5] 3.00-4.00 sec  2.45 MBytes  20.5 Mbits/sec  0.602 ms  0/1773 (0%)
[ 5] 4.00-5.00 sec  2.76 MBytes  23.2 Mbits/sec  0.613 ms  0/2000 (0%)
[ 5] 5.00-6.06 sec  2.97 MBytes  23.5 Mbits/sec  5.398 ms  0/2149 (0%)
[ 5] 6.06-7.00 sec  3.00 MBytes  26.8 Mbits/sec  0.517 ms  0/2176 (0%)
[ 5] 7.00-8.00 sec  2.96 MBytes  24.8 Mbits/sec  0.463 ms  0/2143 (0%)
[ 5] 8.00-9.00 sec  2.95 MBytes  24.7 Mbits/sec  0.573 ms  0/2136 (0%)
[ 5] 9.00-10.00 sec  2.90 MBytes  24.3 Mbits/sec  0.594 ms  0/2098 (0%)
-----
[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Total Datagrams
[ 5] 0.00-10.04 sec  52.5 MBytes  43.9 Mbits/sec  0.000 ms  0/38047 (0%) sender
[ 5] 0.00-10.00 sec  24.8 MBytes  20.8 Mbits/sec  0.594 ms  0/17933 (0%) receiver
iperf Done.
etccomm@etccomm-Veriton-Series:~/0-RAN-Testbed-Automation/OpenAirInterface_Testbed$

```

```

-----
Server listening on 5201
-----
Accepted connection from 10.45.0.2, port 54287
[ 5] local 10.45.0.1 port 5201 connected to 10.45.0.2 port 44680
[ ID] Interval      Transfer      Bitrate      Total Datagrams
[ 5] 0.00-1.00 sec  5.24 MBytes  43.9 Mbits/sec  3792
[ 5] 1.00-2.00 sec  5.25 MBytes  44.0 Mbits/sec  3799
[ 5] 2.00-3.00 sec  5.25 MBytes  44.0 Mbits/sec  3801
[ 5] 3.00-4.00 sec  5.21 MBytes  43.6 Mbits/sec  3772
[ 5] 4.00-5.00 sec  5.28 MBytes  44.3 Mbits/sec  3825
[ 5] 5.00-6.00 sec  5.25 MBytes  44.0 Mbits/sec  3799
[ 5] 6.00-7.00 sec  5.24 MBytes  44.0 Mbits/sec  3798
[ 5] 7.00-8.01 sec  5.24 MBytes  43.6 Mbits/sec  3797
[ 5] 8.01-9.00 sec  5.25 MBytes  44.4 Mbits/sec  3799
[ 5] 9.00-10.00 sec  5.24 MBytes  43.9 Mbits/sec  3793
[ 5] 10.00-10.04 sec  102 KBytes  19.2 Mbits/sec  72
-----
[ ID] Interval      Transfer      Bitrate      Jitter      Lost/Total Datagrams
[ 5] 0.00-10.04 sec  52.5 MBytes  43.9 Mbits/sec  0.000 ms  0/38047 (0%) sender
-----
Server listening on 5201
-----

```

Figure 10: 44Mbps UDP downlink Iperf3 test

Test Duration set at 10 secs

Table 5: UDP Downlink Performance (Core → UE)

Target Rate (Mbps)	Sending Bit Rate (Mbps)	Data Transferred (MB)	Receiving Bit Rate (Mbps)	Data Received (MB)	Arrival Time (s)	Loss (%)	Status
20	20	24.2	12.6	24.2	10	0.26	Did not crash
30	30	36.5	12.8	36.5	10	0	Did not crash
40	40	48.2	11.6	48.2	10	3.3	Did not crash
44	44	52.5	20.8	52.5	10	0	Did not crash
45	45	Crashed at 9 seconds					
50	50	Crashed at 9 seconds					

6.3 TCP Throughput Analysis

TCP throughput testing was conducted without specifying a target bitrate, allowing TCP congestion control mechanisms to determine the achievable throughput dynamically.

6.3.1 TCP Uplink Performance

```

etccomm@etccomm-Veriton-Series:~/O-RAN-Testbed-Automation/OpenAirInterface_Testbed$ sudo ip netns exec uel iperf3 -c 10.45.0.1 -B 10.45.0.2
Connecting to host 10.45.0.1, port 5201
[ 5] local 10.45.0.2 port 49849 connected to 10.45.0.1 port 5201
[ ID] Interval      Transfer    Bitrate      Retr  Cwnd
[ 5] 0.00-1.00 sec  250 KBytes  2.04 Mbits/sec  0    36.8 KBytes
[ 5] 1.01-2.00 sec  346 KBytes  2.86 Mbits/sec  0    49.5 KBytes
[ 5] 2.00-3.00 sec  382 KBytes  3.13 Mbits/sec  0    65.0 KBytes
[ 5] 3.00-4.00 sec  573 KBytes  4.69 Mbits/sec  0    87.7 KBytes
[ 5] 4.00-5.00 sec  700 KBytes  5.73 Mbits/sec  0    117 KBytes
[ 5] 5.00-6.00 sec  891 KBytes  7.30 Mbits/sec  0    153 KBytes
[ 5] 6.00-7.00 sec  1.06 MBytes 8.86 Mbits/sec  0    201 KBytes
[ 5] 7.00-8.00 sec  1.43 MBytes 12.0 Mbits/sec  0    260 KBytes
[ 5] 8.00-9.00 sec  1.86 MBytes 15.7 Mbits/sec  0    356 KBytes
[ 5] 9.00-10.00 sec 1.74 MBytes 14.6 Mbits/sec  0    478 KBytes
-----
[ ID] Interval      Transfer    Bitrate      Retr
[ 5] 0.00-10.00 sec 9.16 MBytes 7.68 Mbits/sec  0          sender
[ 5] 0.00-10.35 sec 7.30 MBytes 5.92 Mbits/sec                      receiver

iperf Done.
etccomm@etccomm-Veriton-Series:~/O-RAN-Testbed-Automation/OpenAirInterface_Testbed$

Server listening on 5201
-----
Accepted connection from 10.45.0.2, port 57503
[ 5] local 10.45.0.1 port 5201 connected to 10.45.0.2 port 49849
[ ID] Interval      Transfer    Bitrate
[ 5] 0.00-1.00 sec  94.7 KBytes 776 Kbits/sec
[ 5] 1.00-2.00 sec  246 KBytes 2.01 Mbits/sec
[ 5] 2.00-3.00 sec  320 KBytes 2.62 Mbits/sec
[ 5] 3.00-4.00 sec  460 KBytes 3.77 Mbits/sec
[ 5] 4.00-5.00 sec  591 KBytes 4.84 Mbits/sec
[ 5] 5.00-6.00 sec  711 KBytes 5.83 Mbits/sec
[ 5] 6.00-7.00 sec  936 KBytes 7.67 Mbits/sec
[ 5] 7.00-8.00 sec  1.10 MBytes 9.18 Mbits/sec
[ 5] 8.00-9.00 sec  1.23 MBytes 10.3 Mbits/sec
[ 5] 9.00-10.00 sec 1.24 MBytes 10.4 Mbits/sec
[ 5] 10.00-10.35 sec 465 KBytes 10.8 Mbits/sec
-----
[ ID] Interval      Transfer    Bitrate
[ 5] 0.00-10.35 sec 7.30 MBytes 5.92 Mbits/sec                      receiver

Server listening on 5201
-----

```

Figure 11: TCP uplink Iperf3 test

The TCP uplink throughput was significantly lower than UDP uplink throughput. However, zero retransmissions were recorded, indicating that the uplink path remained highly reliable

despite limited throughput.

6.3.2 TCP Downlink Performance

```

etccomm@etccomm-Veriton-Series:~/0-RAN-Testbed-Automation/OpenAirInterface_Testbed$ sudo ip netns exec ue1 iperf3 -c 10.45.0.1 -B 10.45.0.2
Connecting to host 10.45.0.1, port 5201
[ 5] local 10.45.0.2 port 49849 connected to 10.45.0.1 port 5201
[ ID] Interval      Transfer    Bitrate    Retr  Cwnd
[ 5] 0.00-1.01 sec  250 KBytes  2.04 Mbits/sec  0    36.8 KBytes
[ 5] 1.01-2.00 sec  346 KBytes  2.86 Mbits/sec  0    49.5 KBytes
[ 5] 2.00-3.00 sec  382 KBytes  3.13 Mbits/sec  0    65.0 KBytes
[ 5] 3.00-4.00 sec  573 KBytes  4.69 Mbits/sec  0    87.7 KBytes
[ 5] 4.00-5.00 sec  700 KBytes  5.73 Mbits/sec  0   117 KBytes
[ 5] 5.00-6.00 sec  891 KBytes  7.30 Mbits/sec  0   153 KBytes
[ 5] 6.00-7.00 sec  1.06 MBytes  8.86 Mbits/sec  0   201 KBytes
[ 5] 7.00-8.00 sec  1.43 MBytes 12.0 Mbits/sec  0   260 KBytes
[ 5] 8.00-9.00 sec  1.86 MBytes 15.7 Mbits/sec  0   356 KBytes
[ 5] 9.00-10.00 sec 1.74 MBytes 14.6 Mbits/sec  0   478 KBytes
-----
[ ID] Interval      Transfer    Bitrate    Retr  sender
[ 5] 0.00-10.00 sec  9.16 MBytes  7.68 Mbits/sec  0
[ 5] 0.00-10.35 sec  7.30 MBytes  5.92 Mbits/sec
iperf Done.
etccomm@etccomm-Veriton-Series:~/0-RAN-Testbed-Automation/OpenAirInterface_Testbed$

```

```

Server listening on 5201
-----
Accepted connection from 10.45.0.2, port 57499
[ 5] local 10.45.0.1 port 5201 connected to 10.45.0.2 port 37297
[ ID] Interval      Transfer    Bitrate    Retr  Cwnd
[ 5] 0.00-1.00 sec  1.46 MBytes 12.3 Mbits/sec  0    205 KBytes
[ 5] 1.00-2.00 sec  445 KBytes  3.63 Mbits/sec  0    239 KBytes
[ 5] 2.00-3.00 sec  1.06 MBytes  8.89 Mbits/sec  0    284 KBytes
[ 5] 3.00-4.00 sec  1.93 MBytes 16.2 Mbits/sec  0    345 KBytes
[ 5] 4.00-5.00 sec  1.55 MBytes 13.0 Mbits/sec  0    417 KBytes
[ 5] 5.00-6.00 sec  1.93 MBytes 16.2 Mbits/sec  0    505 KBytes
[ 5] 6.00-7.00 sec  3.42 MBytes 28.7 Mbits/sec  0    621 KBytes
[ 5] 7.00-8.00 sec  2.50 MBytes 21.0 Mbits/sec  23   496 KBytes
[ 5] 8.00-9.00 sec  2.50 MBytes 21.0 Mbits/sec  0   568 KBytes
[ 5] 9.00-10.00 sec  3.75 MBytes 31.5 Mbits/sec  0   619 KBytes
-----
[ ID] Interval      Transfer    Bitrate    Retr
[ 5] 0.00-10.05 sec  20.5 MBytes 17.1 Mbits/sec  23
-----
Server listening on 5201
-----

```

Figure 12: TCP downlink Iperf3 test

TCP downlink throughput remained relatively close to the UDP downlink throughput values. However, 23 retransmissions were recorded, indicating occasional packet drops in the downlink direction under higher traffic loads.

Performance Analysis:

Table 6: TCP Performance Analysis

Direction	Sender Bit Rate (Mbps)	Receiver Bit Rate (Mbps)	Retransmissions
Uplink	7.68	5.92	0
Downlink	17.1	14.4	23

6.4 Ping and Latency Analysis

ICMP ping testing was conducted using 20 packets transmitted from the UE namespace to the 5G Core Network.

```
etccomm@etccomm-Veriton-Series:~/0-RAN-Testbed-Automation/OpenAirInterface_Testbed$ sudo ip netns exec ue1 ping 10.45.0.1 -c 20
PING 10.45.0.1 (10.45.0.1) 56(84) bytes of data:
64 bytes from 10.45.0.1: icmp_seq=1 ttl=64 time=1093 ms
64 bytes from 10.45.0.1: icmp_seq=2 ttl=64 time=178 ms
64 bytes from 10.45.0.1: icmp_seq=3 ttl=64 time=110 ms
64 bytes from 10.45.0.1: icmp_seq=4 ttl=64 time=99.4 ms
64 bytes from 10.45.0.1: icmp_seq=5 ttl=64 time=1078 ms
64 bytes from 10.45.0.1: icmp_seq=6 ttl=64 time=52.5 ms
64 bytes from 10.45.0.1: icmp_seq=7 ttl=64 time=128 ms
64 bytes from 10.45.0.1: icmp_seq=8 ttl=64 time=130 ms
64 bytes from 10.45.0.1: icmp_seq=9 ttl=64 time=470 ms
64 bytes from 10.45.0.1: icmp_seq=10 ttl=64 time=96.1 ms
64 bytes from 10.45.0.1: icmp_seq=11 ttl=64 time=109 ms
64 bytes from 10.45.0.1: icmp_seq=12 ttl=64 time=105 ms
64 bytes from 10.45.0.1: icmp_seq=13 ttl=64 time=1031 ms
64 bytes from 10.45.0.1: icmp_seq=14 ttl=64 time=60.0 ms
64 bytes from 10.45.0.1: icmp_seq=15 ttl=64 time=80.0 ms
64 bytes from 10.45.0.1: icmp_seq=16 ttl=64 time=79.5 ms
64 bytes from 10.45.0.1: icmp_seq=17 ttl=64 time=1047 ms
64 bytes from 10.45.0.1: icmp_seq=18 ttl=64 time=54.6 ms
64 bytes from 10.45.0.1: icmp_seq=19 ttl=64 time=118 ms
64 bytes from 10.45.0.1: icmp_seq=20 ttl=64 time=107 ms

--- 10.45.0.1 ping statistics ---
20 packets transmitted, 20 received, 0% packet loss, time 19122ms
rtt min/avg/max/mdev = 52.497/311.356/1093.003/384.968 ms, pipe 2
etccomm@etccomm-Veriton-Series:~/0-RAN-Testbed-Automation/OpenAirInterface_Testbed$
```

Figure 13: Ping test

Table 7: Ping and Latency Analysis

Metric	Value
Packets Transmitted	20
Packets Received	20
Packet Loss	0%
Minimum RTT	52.5 ms
Average RTT	311.35 ms
Maximum RTT	1093 ms
Mean Deviation (mdev)	384.96 ms

The test demonstrated zero packet loss, confirming successful end-to-end connectivity. However, substantial latency variation and jitter were observed.

6.5 Parallel Stream TCP Analysis

A TCP throughput test using four parallel streams was performed to evaluate concurrent traffic handling capability.

```

etccomm@etccomm-Veriton-Series:~/0-RAN-Testbed-Automation/OpenAirInterface_Testbed$ sudo ip netns exec ue1 iperf3 -c 10.45.0.1 -B 10.45.0.2 -P 4
Connecting to host 10.45.0.1, port 5201
[ 5] local 10.45.0.2 port 33291 connected to 10.45.0.1 port 5201
[ 7] local 10.45.0.2 port 58067 connected to 10.45.0.1 port 5201
[ 9] local 10.45.0.2 port 46155 connected to 10.45.0.1 port 5201
[11] local 10.45.0.2 port 45475 connected to 10.45.0.1 port 5201
[ ID] Interval      Transfer    Bitrate    Retr  Cwnd
[ 5]  0.00-1.00    sec    211 KBytes  1.73 Mbits/sec    0   35.4 KBytes
[ 7]  0.00-1.00    sec    109 KBytes  892 Kbits/sec    0   33.9 KBytes
[ 9]  0.00-1.00    sec    247 KBytes  2.03 Mbits/sec    0   33.9 KBytes
[11]  0.00-1.00    sec    211 KBytes  1.73 Mbits/sec    0   33.9 KBytes
[SUM] 0.00-1.00    sec    778 KBytes  6.37 Mbits/sec    0
-----
[ 5]  1.00-2.00    sec    96.2 KBytes  788 Kbits/sec    0   39.6 KBytes
[ 7]  1.00-2.00    sec    218 KBytes  1.78 Mbits/sec    0   39.6 KBytes
[ 9]  1.00-2.00    sec    91.9 KBytes  753 Kbits/sec    0   38.2 KBytes
[11]  1.00-2.00    sec    96.2 KBytes  788 Kbits/sec    0   38.2 KBytes
[SUM] 1.00-2.00    sec    502 KBytes  4.11 Mbits/sec    0
-----
[ 5]  7.00-8.00    sec    445 KBytes  3.65 Mbits/sec    0   233 KBytes
[ 7]  7.00-8.00    sec    382 KBytes  3.13 Mbits/sec    0   214 KBytes
[ 9]  7.00-8.00    sec    382 KBytes  3.13 Mbits/sec    0   219 KBytes
[11]  7.00-8.00    sec    445 KBytes  3.65 Mbits/sec    0   218 KBytes
[SUM] 7.00-8.00    sec    1.62 MBytes  13.5 Mbits/sec    0
-----
[ 5]  8.00-9.00    sec    636 KBytes  5.21 Mbits/sec    0   322 KBytes
[ 7]  8.00-9.00    sec    1.12 MBytes  9.39 Mbits/sec    0   310 KBytes
[ 9]  8.00-9.00    sec    445 KBytes  3.65 Mbits/sec    0   276 KBytes
[11]  8.00-9.00    sec    509 KBytes  4.17 Mbits/sec    0   286 KBytes
[SUM] 8.00-9.00    sec    2.67 MBytes  22.4 Mbits/sec    0
-----
[ 5]  9.00-10.00   sec    764 KBytes  6.26 Mbits/sec    0   417 KBytes
[ 7]  9.00-10.00   sec    827 KBytes  6.78 Mbits/sec    0   423 KBytes
[ 9]  9.00-10.00   sec    573 KBytes  4.69 Mbits/sec    0   358 KBytes
[11]  9.00-10.00   sec    636 KBytes  5.21 Mbits/sec    0   372 KBytes
[SUM] 9.00-10.00   sec    2.73 MBytes  22.9 Mbits/sec    0
-----
[ ID] Interval      Transfer    Bitrate    Retr
[ 5]  0.00-10.00   sec    3.90 MBytes  3.27 Mbits/sec    0      sender
[ 5]  0.00-11.31   sec    2.57 MBytes  1.90 Mbits/sec    0      receiver
[ 7]  0.00-10.00   sec    4.11 MBytes  3.45 Mbits/sec    0      sender
[ 7]  0.00-11.31   sec    2.52 MBytes  1.87 Mbits/sec    0      receiver
[ 9]  0.00-10.00   sec    3.13 MBytes  2.62 Mbits/sec    0      sender
[ 9]  0.00-11.31   sec    2.23 MBytes  1.65 Mbits/sec    0      receiver
[11]  0.00-10.00   sec    3.28 MBytes  2.75 Mbits/sec    0      sender
[11]  0.00-11.31   sec    2.27 MBytes  1.68 Mbits/sec    0      receiver
[SUM] 0.00-10.00   sec    14.4 MBytes  12.1 Mbits/sec    0      sender
[SUM] 0.00-11.31   sec    9.59 MBytes  7.11 Mbits/sec    0      receiver

iperf Done.

```

Figure 14: Parallel Stream TCP test

Table 8: Parallel TCP Stream Analysis

Metric	Value
Total Sender Throughput	12.1 Mbps
Total Receiver Throughput	7.11 Mbps
Per Stream (Average)	3 Mbps
Retransmissions	0

6.6 Session Stability Analysis

Extended duration UDP testing was conducted at a target bitrate of 20 Mbps over approximately 100 seconds to evaluate long-term session stability.

Table 9: Stability Test Analysis for 20 Mbps Traffic

Time Duration	Received Bit Rate	Packet Loss	Status
0–58 sec	~10 Mbps	0%	Connected
58 sec	–	–	Client Crash (Control Socket Closed)
58–75 sec	~9 Mbps	0%	Connected
75–99 sec	~3–8 Mbps	43–81%	Massive Packet Loss
99 sec	–	–	Server Crash (Bad File Descriptor)

The system initially maintained stable throughput with zero packet loss. However, prolonged execution caused severe degradation and eventual server failure.

7 Discussion

In UDP performance analysis, the uplink connectivity Table 4 shows that the core network consistently receives a data rate of 8–9 Mbps regardless of how much is sent, whereas the downlink connectivity Table 5 shows that the UE receives a data rate of approximately 12–20 Mbps, which is better than uplink. This can be attributed to two factors:

- i) **TDD Slot Asymmetry:** As defined in 3GPP TS 38.213 V17.1.0, Table 11.1.1-1 [9], TDD Slot Format 34 was employed in the gNB configuration, consisting of 7 downlink slots, 1 flexible slot, and 2 uplink slots per 5 ms transmission period. This results in an inherent DL:UL time ratio of approximately 3.5:1, whereby uplink transmissions are allocated significantly fewer time resources than downlink transmissions. Consequently, the maximum achievable uplink throughput is structurally constrained by the slot configuration.
- ii) A maximum stable **UDP throughput of 44 Mbps** was observed during incremental bitrate testing, whereas at **45 Mbps and above**, the iperf3 control socket terminated unexpectedly, causing connection failure. This limitation is attributed to computational bottlenecks in the software-based RFSim environment, where the UE, gNB, and 5G Core execute simultaneously on a **general-purpose processor (GPP)**. At higher bitrates, increased computational demand associated with **PHY-layer processing, packet handling, and RF channel emulation** likely exceeded available CPU resources, resulting in degraded real-time performance. Similar CPU-related throughput limitations in OAI-based deployments were reported by **Lin and Huang [13]**, where CPU utilization increased substantially with transmission bandwidth.

In TCP performance analysis, Table 6 shows that **TCP uplink throughput** was significantly lower than **UDP uplink throughput**, with TCP achieving only **5.92 Mbps** compared to approximately **8–9 Mbps** for UDP. In contrast, **TCP downlink throughput (14.4 Mbps)** remained comparable to **UDP downlink throughput (12–20 Mbps)**. Uplink transmission exhibited **zero retransmissions and packet loss**, indicating a reliable but delayed communication path. However, **23 retransmissions** were observed during downlink transmission, consistent

with the **3.3% UDP packet loss** observed at higher bitrates, indicating occasional packet drops in the downlink path. Furthermore, in UDP Tables 4 and 5, uplink receiver time was always much longer than 10 seconds (23.96, 34.66, 44.92 seconds), whereas downlink transmission completed within the expected duration of 10 seconds. This demonstrates that **uplink suffers buffering delay, while downlink does not**.

The higher reliability of the uplink is attributed to the **TDD slot configuration**, which allocates fewer uplink transmission opportunities, thereby reducing packet arrival rates and preventing receiver buffer overflow. In contrast, the higher downlink slot allocation increased packet arrival rates at the UE software receiver, occasionally exceeding buffer capacity and resulting in packet loss and retransmissions.

In Table 7, a **ping test of 20 ICMP packets** recorded **zero packet loss**, indicating a stable communication path. However, significant latency variation was observed, with an average **RTT of 311.35 ms** and a **mean deviation of 384.96 ms**, indicating severe jitter and inconsistent delay performance. This behavior is attributed to the software-based implementation of the testbed, where **RF channel emulation, packet processing, and 5G stack execution** are performed on a **general-purpose processor (GPP)**. Variations in CPU scheduling and processing overhead likely introduced non-deterministic delays, resulting in high and fluctuating RTT values. Similar computational overhead in OAI-based deployments was reported by **Lin and Huang [13]**, where CPU utilization increased significantly with system load, highlighting the sensitivity of software-defined RAN implementations to processing constraints.

In Table 8, the abrupt degradation observed during the **20 Mbps stability test**, characterized by unexpected **control socket termination**, severe packet loss (**43–81%**), and eventual **server failure**, indicates instability under sustained traffic conditions. Rather than gradual performance degradation, the failure pattern suggests **resource exhaustion within the software-based OAI/RFSim environment**, potentially caused by cumulative CPU scheduling overhead, buffer saturation, or socket management limitations.

In Table 9, the use of **four parallel TCP streams** increased the aggregate throughput to **12.1 Mbps**, exceeding the performance of a single TCP stream (**7.68 Mbps**). Bandwidth was distributed relatively evenly among the streams, indicating effective operation of the **TCP congestion control and fairness mechanisms**. These results suggest that the available **GTP tunnel capacity** can be utilized more efficiently through multiple concurrent TCP connections, thereby improving overall throughput performance.

8 Challenges and Limitations

Despite successful deployment and integration of the **O-RAN compliant 5G Standalone testbed**, several challenges and limitations were observed. Since the implementation relied on the OpenAirInterface RF Simulator (RFSim), all major network components — including the UE, gNB, 5G Core, and Near-RT RIC — were executed simultaneously on Commercial Off-The-Shelf (COTS) hardware using a **general-purpose processor (GPP)**. This introduced significant computational overhead, resulting in throughput limitations, high latency, severe jitter, and instability under sustained traffic conditions. A maximum stable **UDP throughput of 44 Mbps** was observed, beyond which connection failures occurred due to CPU resource saturation and software processing bottlenecks. Furthermore, the software-based nature of RFSim limits the ability to evaluate realistic radio channel effects such as interference, fading, and hardware impairments present in practical wireless environments.